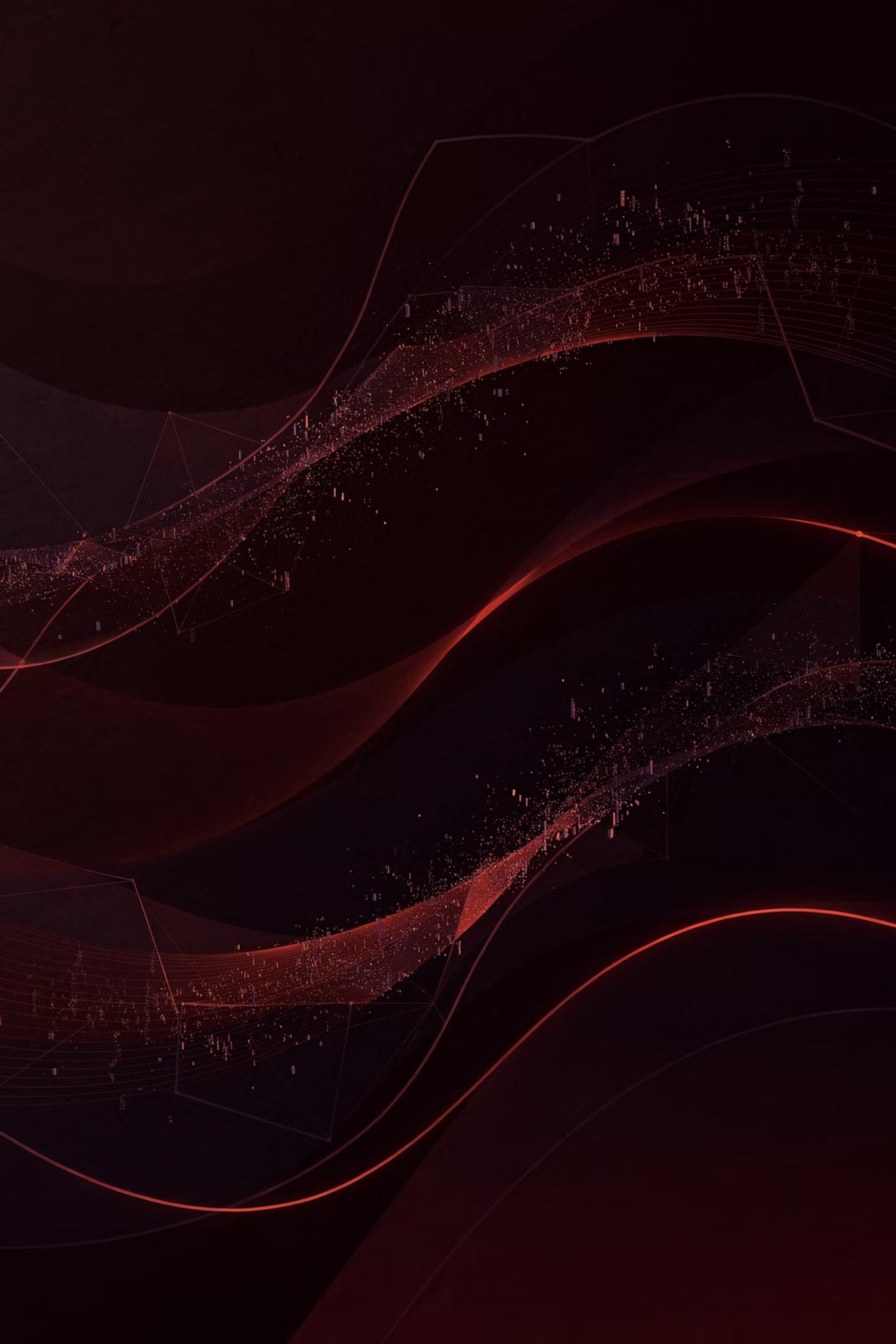


The background of the slide is split. The left half features a dark, abstract digital landscape with glowing orange and red wavy lines, particle trails, and geometric wireframe structures. The right half is a solid, deep maroon color.

HDD vs SSD in Digital Forensics

Interesting aspects

01

Architectural Principles

LBA mapping, FTL indirection, and wear levelling

03

Acquisition & Integrity

Write blockers, hash verification, deleted data recovery

02

The Three Critical SSD Mechanisms

TRIM variants, autonomous garbage collection, physical fragmentation

04

Advanced Techniques & Encryption

Factory access mode, chip-off, SEDs, and chain of custody

Architectural Principles



HDD: Sequential, Stable Addressing

An HDD maps each Logical Block Address (LBA) to a **fixed, stable physical location** on a rotating platter. No relocation occurs without explicit OS involvement — the physical location of a data block is deterministic and persistent over time.

SSD: The FTL Indirection Layer

The **Flash Translation Layer (FTL)** — internal firmware invisible to the OS — maps logical addresses onto NAND cells. Wear-levelling algorithms continuously migrate data between cells, breaking the classical forensic link between a logical file address and any static physical location.

Three Critical SSD Mechanisms

Three technologies intrinsic to SSD operation may autonomously alter evidence after seizure.



TRIM

Issued by the OS upon file deletion or format. Triggers garbage collection - and critically, continues running even after power-off is resumed.



Autonomous Garbage Collection

The SSD microcontroller performs read/write operations independently of the host. Internal writes occur even when no commands are received from the PC.



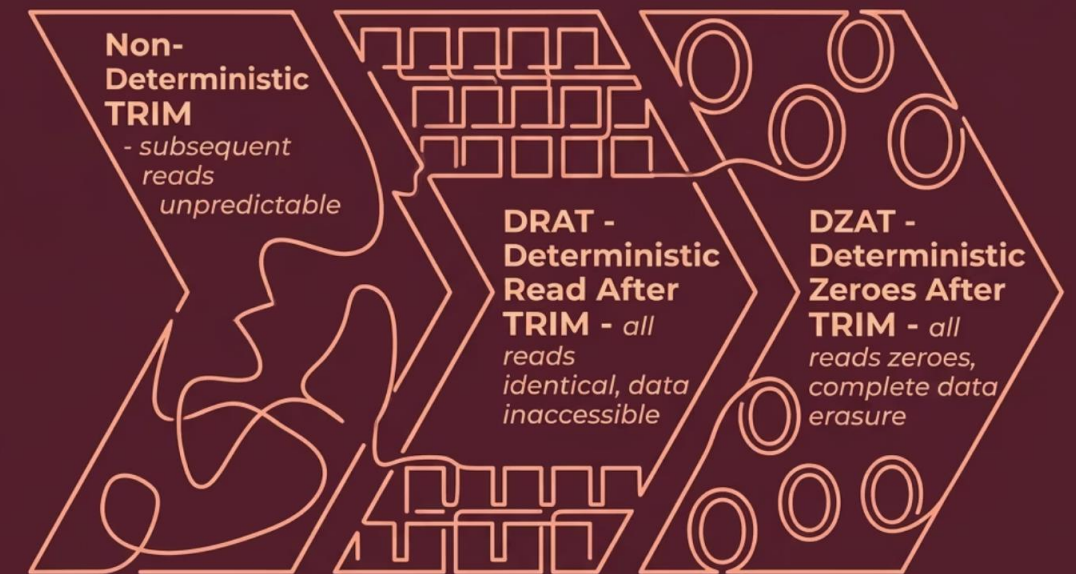
Wear Levelling

Continuously relocates data to prevent cell fatigue. Buggy firmware in some drive models produces forensically unpredictable behaviour highly dependent on the specific model.

TRIM Variants: What Happens After Deletion

The SATA standard defines three distinct post-TRIM read behaviours. Understanding which variant is implemented on the target drive is relevant before any acquisition attempt.

- ❏ With **DRAT** and **DZAT**, access to real data is effectively prevented *immediately* after the TRIM command is issued - even following a Quick Format.



The Write Blocker Problem

Hardware write blockers are the gold-standard forensic acquisition tool - but their effectiveness is fundamentally compromised on SSDs.

The Core Limitation

A write blocker intercepts commands from the host. The SSD's internal microcontroller operates below that interface - garbage collection and wear levelling proceed regardless of any externally applied write block.

TRIM Is a Separate Command

Some forensic-grade write blockers that advertise write protection nonetheless **fail to block TRIM**. This can be experimentally verified using `blkdiscard` on Linux after connecting the drive through a write blocker.

Accepted Consequence

Background processes that cannot be halted by traditional write blockers are now considered an accepted limitation in SSD forensic practice - not an error by the examiner.

Hash Integrity: Logical Stability vs Physical Flux

The Debate

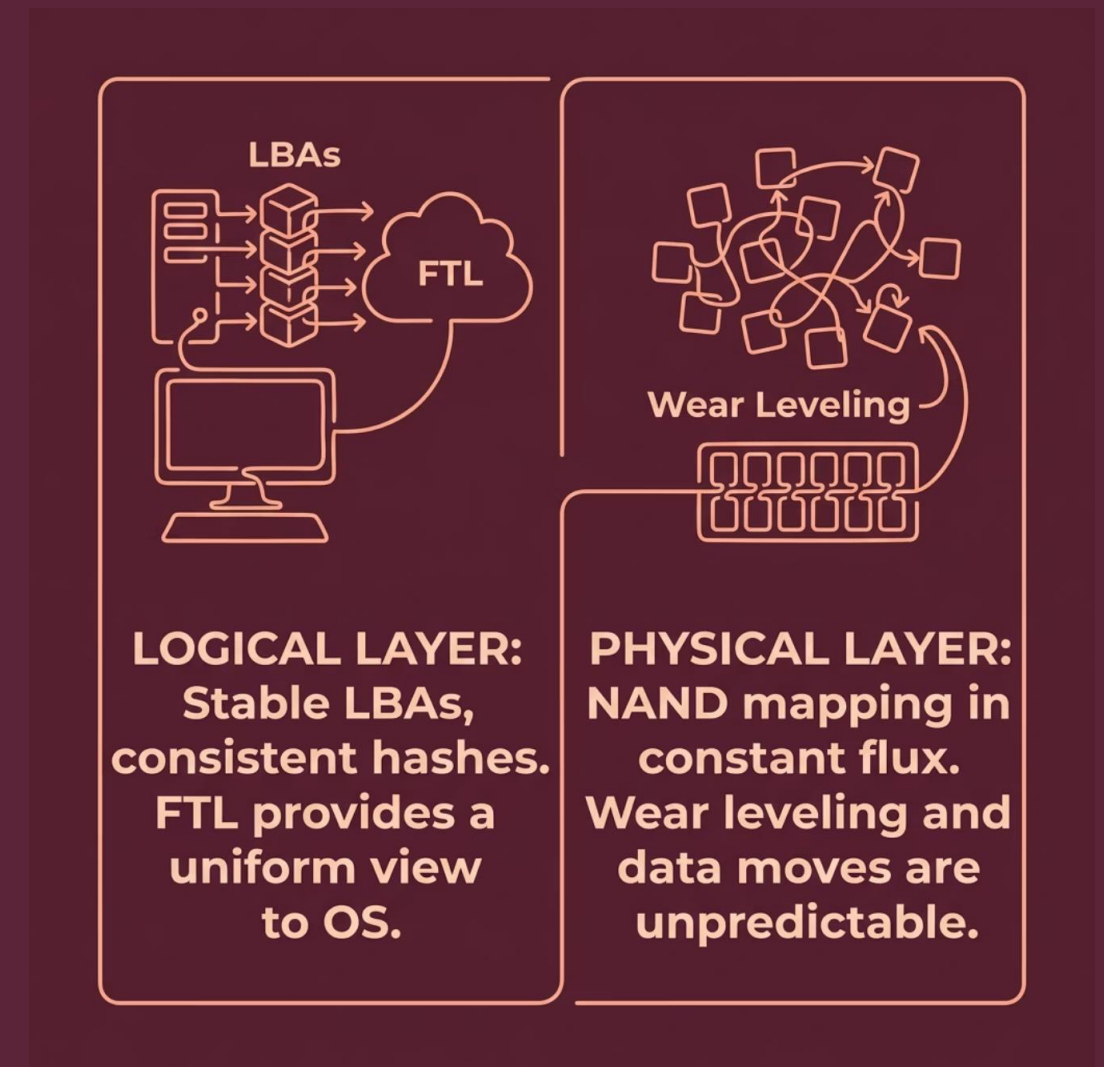
It has been argued that hashing an SSD cannot confirm integrity, because data changes over time. Experimental evidence tells a more nuanced story.

Empirical Finding

In repeated testing, hash values computed over the same drive did not change between sessions. The FTL presents a consistent logical view to the OS across reads.

Relevant Distinction

Hash consistency confirms logical-layer stability only. Internally, physical NAND mappings may be in constant flux — the same LBAs return the same data, but the underlying cells may be entirely different.



Deleted Data Recovery Contrast

HDD — Highly Predictable

Usual file deletion removes the index pointer. The underlying data remains at its fixed physical location indefinitely, destroyed only when overwritten. File carving techniques are standardised and reliable.

SSD — Highly Variable

With TRIM enabled: 0% of deleted files recovered in controlled studies. With TRIM disabled: up to 100% recovery. Even a Quick Format can render an SSD's entire content unrecoverable within minutes — a capability unavailable on non-encrypted HDDs.

- ❏ On SSDs, only *existing* data can be assumed accessible. Retrieval of deleted or formatted data is not guaranteed under any circumstances.

Advanced SSD Acquisition Techniques



Factory Access Mode

Temporarily disables controller background processes (GC, wear levelling, address remapping). Enables access to non-addressable blocks including the over-provisioned area. Implemented for select controller families in tools such as **PC-3000** (Ace Laboratory).



Chip-Off Forensics

NAND chips are physically desoldered and read via a hardware programmer. Highly complex: data is typically encrypted by the controller, wear-levelling maps must be reverse-engineered, and the process is destructive and irreversible.



Over-Provisioning Area

SSDs carry more physical NAND than their advertised capacity (e.g. a 120 GB drive may hold 128+ GB physically). The **OP area** can retain recently deleted data before garbage collection processes those blocks -accessible only via factory mode or chip-off.



Encryption & Self-Encrypting Drives

TCG OPAL & AES-256 Hardware Encryption

Modern SSDs frequently implement hardware-level encryption. A crypto erase destroys the internal encryption key in under one second — rendering every NAND block unreadable instantaneously, without overwriting a single cell.

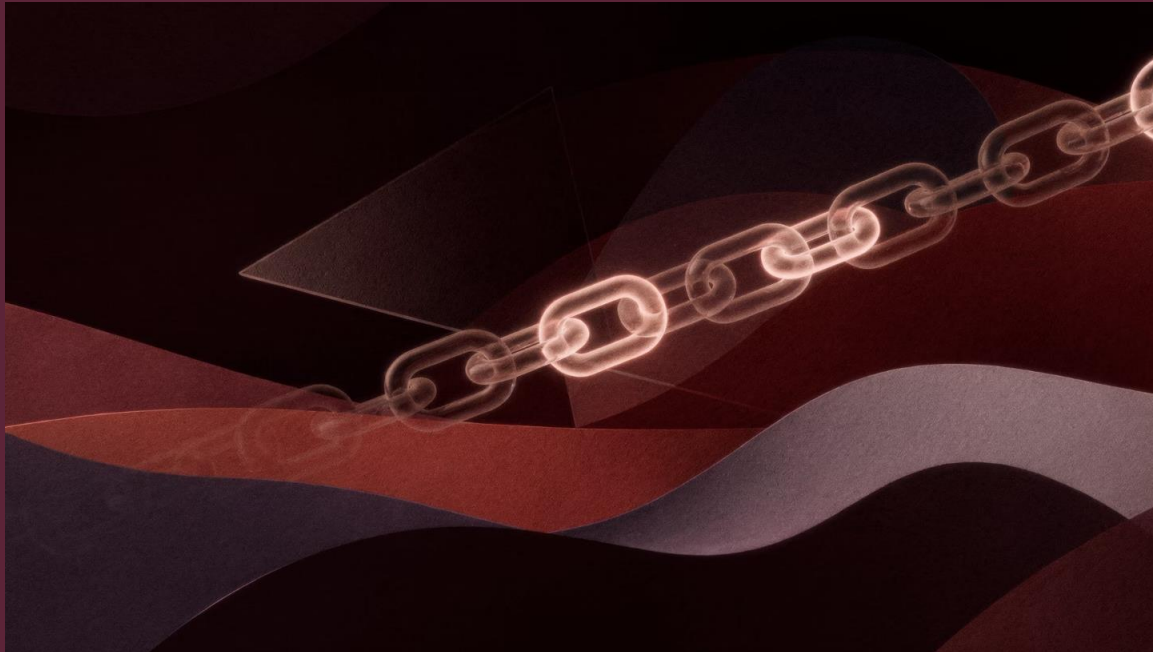
Forensic Implications

- A suspect with access to a SED can irreversibly destroy all evidence with a single command
- No chip-off technique recovers data — the key no longer exists
- HDDs without overlying FDE software do not present this native risk
- NVMe drives require NVMe-specific commands (`nvme sanitize` via `nvme-cli`) — ATA Secure Erase is ineffective

Forensic Comparison: HDD vs SSD

Forensic Dimension	HDD	SSD
Deleted data recovery	High — reliable file carving	Variable; often 0% with TRIM active
Evidence stability	High — no autonomous processes	Low — GC and wear levelling run in background
Write blocker effectiveness	Complete	Partial — does not block internal GC or TRIM
Hash integrity over time	Guaranteed	Guaranteed at logical level only
Standard acquisition	Bit-by-bit image via SATA	Requires attention to TRIM, GC, OP area
Rapid evidence destruction	Difficult without dedicated tools	Trivial — TRIM + GC or crypto erase
Chip-off utility	High — data at fixed locations	Low — FTL must be reverse-engineered
Drive encryption risk	N/A by default	Crypto erase = instant destruction
Manufacturer dependency	Low — consolidated standards	High — firmware, controller, OP area vary

Chain of Custody Challenge



The SSD Paradox

With an HDD, accidental post-seizure alteration is implausible and easy to argue against. With an SSD, the drive **can change its own content autonomously** — without any intervention by the investigator — after power is restored.

Documentation Requirements

To preserve admissibility, forensic examiners must precisely document at time of seizure:

- TRIM flags and TRIM behaviour variant (Non-deterministic / DRAT / DZAT)
- Over-provisioning status and OP area accessibility
- Presence and state of hardware encryption (SED / TCG OPAL)
- Firmware version and controller family

Interesting thoughts

Write blockers are necessary but insufficient for SSDs

Internal controller processes continue regardless. Document this limitation explicitly in your acquisition report.

TRIM status is the first variable to establish

Recovery prospects collapse to near zero on TRIM-enabled drives. Determine TRIM variant before any acquisition strategy is chosen.

Hash consistency is logical, not physical

A matching hash confirms the OS-visible data has not changed — it does not confirm NAND-level stability.

Encryption can make evidence destruction trivial

A SED crypto erase takes under one second. Identify hardware encryption at seizure and treat it as an immediate risk vector.

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